

SkillClaw

Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

SkillClaw enables LLM agent skills to continuously evolve by aggregating cross-user interaction trajectories and applying agentic, evidence-driven skill updates.

Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently — No System-Level Learning

STATIC SKILL PARADIGM

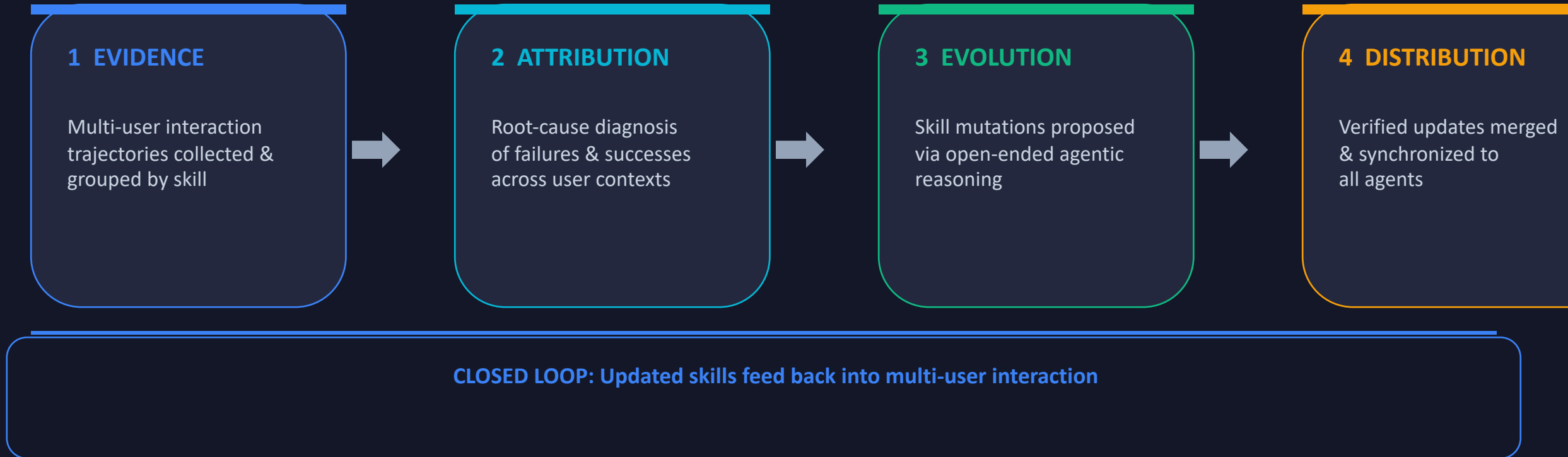
- Skills manually installed from centralized hub
- Solutions discovered during interaction never persist beyond individual sessions
- Same patterns of failure and recovery repeatedly observed across users
- Each user forced to rediscover solutions independently
- Memory-based methods: store past trajectories but remain tied to specific instances
- Skill-based methods: treat library as static resource that does not evolve



EVOLVING SKILL PARADIGM (SkillClaw)

- Skills continuously updated from runtime interaction
- Cross-user trajectories aggregated as shared evidence
- Agentic evolver analyzes patterns, diagnoses failures, proposes mutations
- Conservative verification prevents regressions
- Updated skills synchronized back to all agents automatically
- Knowledge accumulates at the system level

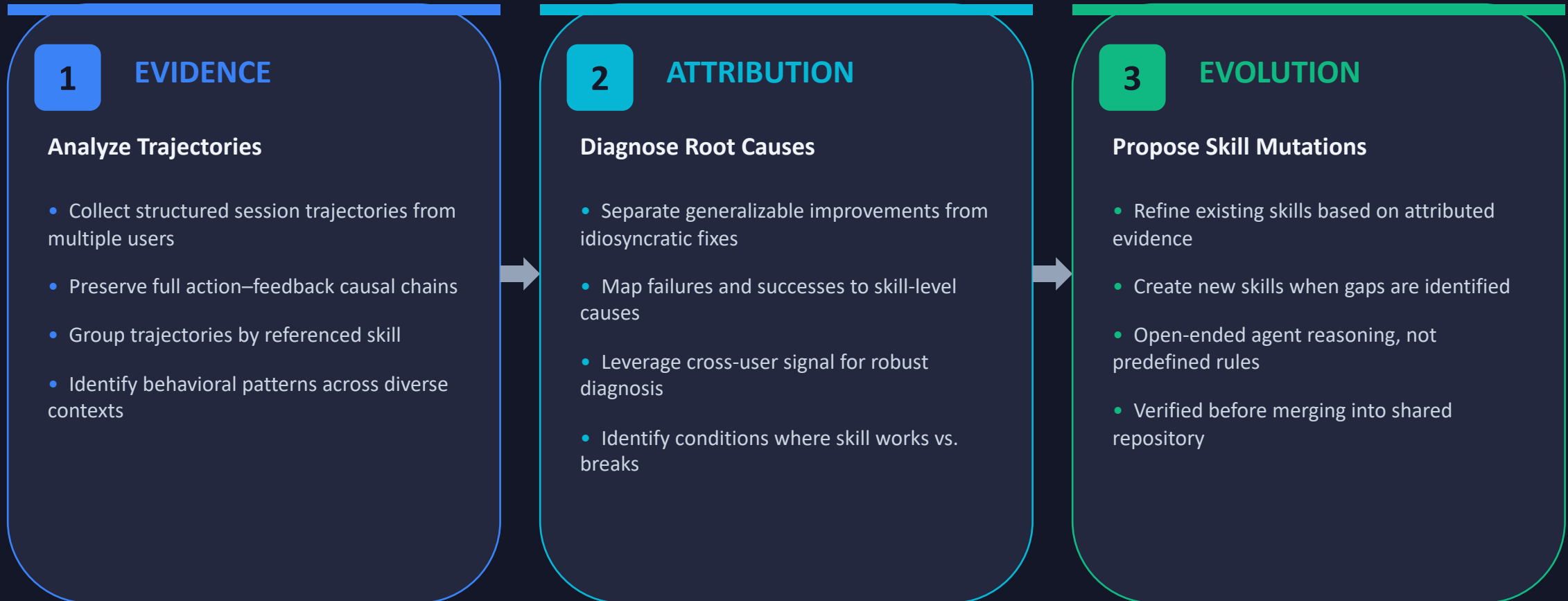
Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution



Independent agents produce structured session trajectories preserving full action–feedback causal chains. Trajectories are aggregated across users and grouped by referenced skills, forming a shared evidence base. The evolution process is driven by an autonomous agent performing open-ended reasoning — not predefined update rules.

Agentic Evolver: Evidence → Attribution → Evolution

Three-stage pipeline for context-aware skill mutations through open-ended reasoning



The evolver is an autonomous agent — it performs open-ended reasoning over interaction evidence, not predefined update rules.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary, revealing both conditions under which it works and those under which it breaks. A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

COMPATIBLE CLAW-STYLE AGENT SYSTEMS

OpenClaw

CoPaw

ZeroClaw

NanoClaw

AGGREGATION MECHANISM

IronClaw

User A
(Context α)



NemoClaw

User B
(Context β)



User C
(Context γ)



Shared
Evidence
Base



Evolved
Skill

Evaluation on WildClawBench using Qwen3-Max as the backbone model demonstrates substantial improvements across tasks.

Case Study: Multimodal Creative Pipeline — 6 Days of

1 of 6 ACCEPTED

Day	Candidate Skill	Skill Function	Outcome	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & env setup	ACCEPTED	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	REJECTED	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (PDF/video/image)	REJECTED	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual gen, validation	REJECTED	Keep current best pool
5	multimodal-creative-task-pipeline + validate-required-input-files	Creative pipeline + per-file fail-fast validation	REJECTED	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / doc-to-visual paths	REJECTED	Not admitted to next cycle

Conservative verification in action: Only 1 of 6 candidates passed validation over 6 days.
The system prevents regressions by maintaining the current best pool when new candidates do not improve.

Case Study: Git Push Auth-Fallback — Iterative Refinement

Days

3 of 6 ACCEPTED

Day	Candidate Skill	Change Summary	Outcome	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	ACCEPTED	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	ACCEPTED	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	ACCEPTED	Keep current best pool
4	(none — retest)	Same-pool retest; validator confirmed current pool as best	ACCEPTED	Continue same best pool
5	git-push-with-auth-fallback	Added 'push hang treated as auth failure'; no improvement	REJECTED	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	REJECTED	Not admitted to next cycle

Iterative refinement succeeds when evidence supports it: 3 accepted updates refined the skill before plateauing. Contrast with creative pipeline (1/6): verification rate reflects evidence quality, not a fixed threshold.

Key Takeaways: Three Properties That Distinguish SkillClaw

1

COLLECTIVE EVOLUTION

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem

Individual user sessions produce trajectories that feed into a shared evidence base. The system learns from all users, not just one — cross-user signals expose behavioral boundaries that a single user rarely generates alone.

2

FULLY AUTOMATIC

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention

No human curation required. The closed loop from interaction to evolution to distribution runs autonomously. Conservative verification ensures only improvements are merged.

3

AGENTIC EVOLUTION PARADIGM

Skill updates are produced through open-ended reasoning rather than predefined update rules

The evolver is an autonomous agent that reasons over evidence, attributes root causes, and proposes context-aware mutations — not a rule engine applying templates.

Implication: SkillClaw demonstrates that multi-user agent ecosystems can be designed to continuously improve through agentic, evidence-driven skill evolution — moving beyond static skill deployment.